

Forecasting film audience ratings: A natural language processing approach to script and production data

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ABSTRACT

The film industry is an important entertainment avenue for audiences of all ages. Demand for good quality scripts remains a core element of this industry, rendering the screenplay a pivotal tool at the green lighting stage.

While previous work addressed isolated elements influencing the performance of a movie, this research aims to bring together known influential factors and some novel approaches by applying Natural Language Processing (NLP) and Machine Learning (ML) techniques to analyse movie scripts, with the aim of extracting valuable insights and patterns that are able to predict the audience rating as collated by the Internet Movie Database (IMDb).

This research helps producers determine which movies are most viable for financing. By providing a sound method to sift through and rank the various script projects presented to them, they can focus on scripts that are likely to perform better.

Methods adopted in this research include the use of lexicons for the extraction of linguistic features, the analysis of emotional arcs in movies, embedding strategies for the script and statistical features generated from sentiment analysis. These features are concatenated to cast and crew specific factors to train various regression models by using a forward rolling window training strategy.

1. Introduction

With thousands of established or aspiring screenwriters, the volume of movie scripts written each year compared with the amount of producers, production companies and available budgets, requires a proper and careful selection of scripts to assess which can be given a green light for production.

Based on experience, movie producers typically predict a script's potential success by finding past movies with similar storylines or content and using how well those films performed financially as a guide to estimate the new movie's likely earnings [1].

The difficulty with assessing the success potential of a movie script at such an early stage lies in the limitation of information that can be used to make such a prediction. Traditionally, important elements such as the cast and crew, music composer, budget, release date, visuals and sound are available after the release of a movie, when most, if not all the budget allocated to it would have been spent.

To ensure a movie's success, several key elements are crucial. The experience of producers, writers, and main actors, along with their past performance, significantly influences a production's potential. The

chosen genre is also vital, as it sets audience expectations and can inherently attract higher viewership for certain types of films. Furthermore, an established director's track record contributes to audience appeal. Ultimately, a high-quality screenplay, one that is original, well-paced, engaging, and features relatable characters, strong dialogue, emotional resonance, and a satisfying resolution, forms the bedrock of a successful movie.

2. Related works

Regression analysis has been used since 1983 to predict movie success [2]. Box-office receipts are frequently used as a measure of performance as Return on Investment (ROI) is a main key performance indicator for investors.

2.1. Box-office prediction

Expected revenue from a movie was analysed in a research paper which classified box-office receipts into one of nine categories, ranging

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from ‘flop’ to a ‘blockbuster’. Neural networks were used on features normally available upon release to achieve an average correct classification rate of 36.9% across the nine categories [3]. Movie revenues were also researched by using features about ‘who’ is in the cast, ‘what’ is the movie about, ‘when’ a movie would be released as well as ‘hybrid’ features such as annual profitability percentage by genres, average genre expertise of cast, keywords and competition in terms of other movies that would be released at the same time. Using a Lasso Regressor the researchers achieved a Root Mean Squared Error (RMSE) of 0.878 [4].

2.2. Audience rating prediction

Other studies focused on the *user rating* of movies. A study of *Twitter* and *YouTube* textual features (comments) on a dataset of 70 movies achieved a RMSE of 0.4459 [5]. In another study published in 2014, the authors used linear combination, multiple linear regression and neural networks on features extracted from IMDb, including genres, directors, actors, writers, country, film locations and runtime, to achieve what they refer to as an average Prediction Absolute Error, which essentially is the Mean Absolute Error, of 0.6973, with 76.8% of the 1462 movies in the test set being predicted within 1 unit of rating [6]. An interesting aspect of this study was the weight that the actors had in their multiple linear regression model. One might argue that at a very early stage of analysis, the actors would not yet have been locked in and hence this feature might not be a realistic feature to use in such a model.

Later in 2020, a study on *user rating* prediction used convolutional neural networks with three dense layers based on historical values, metadata features, categorical features, topical and social features such as director, actor, genres and production companies’ historical average ratings, *Facebook* likes or gross receipts (as may be applicable) together with budgets, duration of the movie, release dates, plot keywords, tagline, content rating and storyline summary, to achieve a Mean Absolute Error of 0.83 by training on a dataset containing 3819 movies released from 2000 until 2013 and testing on a dataset containing 498 movies released after 2013 [7]. This approach yet again made use of features which are likely not to be available at green lighting stage.

2.3. NLP based predictions

Research into predicting movie success using NLP began with a 2006 study that used the simple “bag-of-words” approach, which only analysed movie spoilers, not full scripts [8]. An improvement came in 2010 when researchers extracted features like genre, word counts, and word meanings directly from actual screenplays [9].

Additional work was conducted in 2017, focusing on the analysis of movie scripts as unstructured text [10]. Using 978 movie scripts collected from the Internet Movie Script Database (IMSDB) [11], the authors applied the SentiWordNet [12] vocabulary dictionary to analyse each script into a series of positive and negative scores. Using various smoothing techniques the researchers extracted the highest and lowest positive scores from story progression plots, then employed SimpleKMeans clustering with WEKA to identify two dominant patterns in script narratives.

2.4. Emotion analysis

The use of the Linguistic Inquiry and Word Count (LIWC) analysis tool on movie scripts collated from IMSDB and *SimplyScripts.com* was combined with consumption experience carryover theory to predict movie revenues in a 2021 published study [13]. The authors concatenated 93 LIWC extracted topic features with movie properties such as director power, star power, Motion Picture Association of America (MPAA) rating, whether a movie is a sequel, the genre, year of release and the movie studio involved. Using an Auto-Gaussian spatial model [14] to augment LIWC categories by considering both

observed effects and unobserved correlations between past and new movies, ultimately identifying topic-specific carryover effects and successfully estimating movie revenue by analysing cross-movie similarity and consumer reviews, with their model based on script similarity alone achieving an adjusted R^2 of 0.58.

The use of emotional arcs and the clustering of movie titles based on the shape of such arcs was demonstrated in a research which mapped the emotional trajectory of 6174 movie script subtitles. These clusters were then used to predict the overall success parameters of the movies including box office revenues and viewer satisfaction levels (through IMDb ratings), awards as well as a number of reviews. In this study the authors find that movie stories are dominated by six basic shapes (see Fig. 1) which reflect the work and observations of Kurt Vonnegut [15], with the highest revenues associated with the *Man in a Hole* shape, which resulted in financially successful movies independent of genre type and production budget [16]. While the prospect of using these well defined clusters for predicting user rating was promising, it was found that despite slight differences, mean ratings across the six clusters varied from 6.45 to 6.64 with a standard deviation between 0.94 and 0.99. Such variances may not be significant for creating important features out of these clusters. The authors’ results showed that only the *Man in a Hole* cluster had significantly higher earnings in its home market than the others. A separate analysis also revealed a strong positive link between movies belonging to the *Rags to Riches* group and their IMDb ratings.

3. Methodology

3.1. Sourcing data

The first and most important element of the dataset comprises the scripts or screenplays of movies. Most research conducted on screenplay analysis made use of screenplays sourced from the IMSDB. At time of compilation 1231 screenplays were downloaded from the website. Other scripts were sourced from [18–20] and GitHub repositories [21–25]. A combined dataset of 2785 full length screenplays was collated.

Given the restricted amount of available screenplays it was decided to augment the dataset with subtitles of movies. Subtitles have been used in other research, namely that of [16,17,26,27] with an adequate level of success in clustering similar titles. Subtitles for this paper were sourced from [28–30]. An additional 32,077 titles were added to the full length screenplay dataset, resulting in a total of 34,862 titles before filtering. While subtitles are designed for viewers and might contain the occasional sound queue, screenplays are the blueprint for making the movie and would include additional important information such as scene descriptions, character actions, locations and sometimes even camera directions, hence providing the full narrative context, not just the spoken words. While recognising the potential bias introduced by the introduction of subtitles on the prediction performance of full scripts, our evaluation strategy examines the performance of our model on both full script movies as well as subtitle based movies.

In conjunction with the sourcing of scripts and subtitles, the *meta-data* which would be incorporated in the training data was being collated from multiple sources. Numerous datasets could be identified for such data, mainly those from [31,32]. Any missing data was obtained by linking directly to IMDb using the API provided by the *Python* package *Cinemagoer* [33].

Another important source of direct information was obtained from IMDb [34]. The following metadata features were collated for each movie title: *movie name*, *movie id*, *filename*, *genres*, *rating*, *votes*, *writer*, *producer*, *director*, *year of release*, *actors*.

In deciding what features to include in the training dataset, a balance between early prediction methods, historical statistics and other available factors to include in the training dataset was struck. The underlying rationale was to choose those features which can reasonably be assumed to be available before any major production budget is

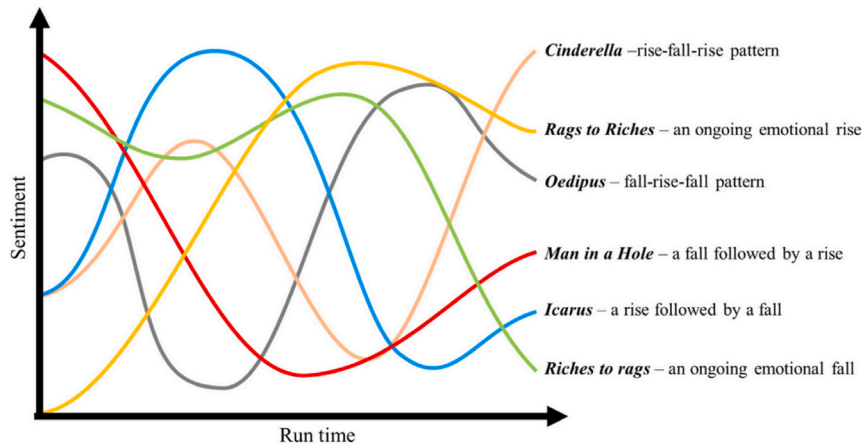


Fig. 1. Six emotional trajectories of movies.
Source: Adapted from [17].

expended. We have decided to include director, writer, producer and actor related factors on the basis that these are the basic cast and crew who will be the first to be secured. For each key crew and cast member identified above we have looked at their past productions and calculated an average rating based on the IMDb rating of the movies they participated in the previous 5 years. It may be argued that the contracting of lead cast and crew occurs after ‘green-lighting’ a movie but as in any other project, the choice of the team can have a significant impact on the results. While some movies with large production budgets and lead actors have under-performed at the box-office, star actors have traditionally attracted better ratings. We account for this inclusion in our evaluation of the model by testing a version that does not incorporate such cast and crew features.

3.2. Linguistic features

The linguistic features of the sourced scripts were analysed using the LIWC tool. The internal dictionary of this tool is composed of over 12,000 words, word stems, phrases and select emoticons. Each entry is allocated to one or more categories, designed to assess various psychosocial constructs [35]. Over 100 dimensions of feature data were extracted for each movie title. As many of the features exhibited multicollinearity when related to the rating value and between themselves, *Recursive Feature Elimination* was employed to retain 115 out of the 151 dimensions to be concatenated to the main metadata. A list of the retained features can be found in [Appendix B](#).

3.3. Emotional arcs

To extract meaningful values from the script, a sentiment analysis tool was employed. We have used the Valence Aware Dictionary and sEntiment Reasoner (VADER) lexicon to extract the *compound* element of the polarity score of the text/script, which is split into sentences. VADER is a sentiment analysis tool that analyses feelings in text using a word list and rules. It is specifically tuned to understand sentiments expressed on social media [36]. The compound score is the overall sentiment measure for a sentence, scaled from -1 (most negative) to +1 (most positive). It is computed by adding up the sentiment scores of the words, adjusting them according to VADER’s rules, and then normalising the total.

Given the different lengths of movie scripts, the raw score results are not homogeneous in length. In addition, plots generated based on the raw values do not provide a clear visual of the arcs. To smooth the results and represent them in a 100 dimensional vector (representing 100% of the length of a movie) we have used Discrete Cosine Transformation (DCT) to obtain smooth emotional arcs, like that obtained in [Fig. 2\(b\)](#).

3.3.1. Clustering emotional arcs

Dynamic Time Warping (DTW), which is a widely used technique for aligning time-dependent sequences by optimally matching their shapes, [37] was applied for the analysis of the shapes of emotional arcs extracted from movie dialogue, where emotions may unfold at different rates across movies.

Having created a distance matrix based on distance time warping we proceeded to cluster the vectors using *KMedoids* [38]. Using the *silhouette coefficient* as the determinant for optimal cluster cohesion, the algorithm achieved the highest silhouette coefficient at 6 clusters. The respective groupings were subsequently plotted, with a mean arc, smoothed using *UnivariateSpline*, plotted for each cluster, to represent the predominant emotional trajectory within that cluster of movies, as shown in [Fig. 3](#).

Research at the University of Vermont’s Computational Story Lab identified six basic story types: *Rags to Riches*, *Riches to Rags*, *Icarus*, *Oedipus*, *Cinderella*, and *Man in a Hole* [39]. However, a clustering analysis in this research, while recognising *Icarus*, *Oedipus*, *Cinderella*, and *Man in a Hole* patterns, found that simple *Rags to Riches* and *Riches to Rags* trajectories were not prominent in the emotional arcs of the dataset. Instead, two new, opposing patterns emerged: *Phantom* ([fall, rise, fall, rise]), characterised by alternating setbacks and recoveries, and *Scarface* ([rise, fall, rise, fall]), which shows initial success followed by a downfall, recovery, and a final decline.

The allocated cluster was used as a feature for the regression model.

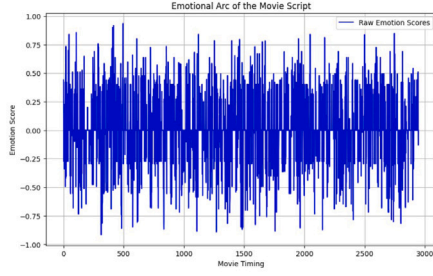
3.3.2. Emotion analysis

Emotions characterise human life. Affective processes are responsible for our behaviour, enacting and sustaining motivation. Research has found connections between affect and human performance, such as creativity and decision making [40]. We have chosen two, lexicon based tools, to analyse the scripts in the dataset. These lexicons have been chosen to calculate intensity and presence scores.

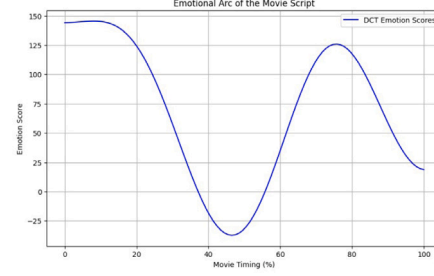
- *The NRC Emotion Intensity Lexicon (NRC-EIL)*: This lexicon is used to calculate the intensity score of the text [41].
- *NRC Word-Emotion Association Lexicon (aka EmoLex)*: This lexicon is used to calculate the presence score of the text [42] and was computed using the Python package *NRCLex* [43].

Using composite emotions derived from *Plutchik’s Wheel of Emotions* [44] we also computed features for *surprise_intensity*, *love_intensity*, *contempt_intensity*, *positive*, *surprise*, *trust*, *fear*, *negative*, *disappointment* and *optimism*.

Having extracted raw and DCT values from the script, we extracted additional statistical features, with the aim of capturing patterns in the development of the emotional arc of each movie.



(a) VADER raw compound values for the movie *A Few Good Men*. Script Source: [11]



(b) DCT values for the movie *A Few Good Men*. Script Source: [11]

Fig. 2. Constructing the Emotional Arc for movies.

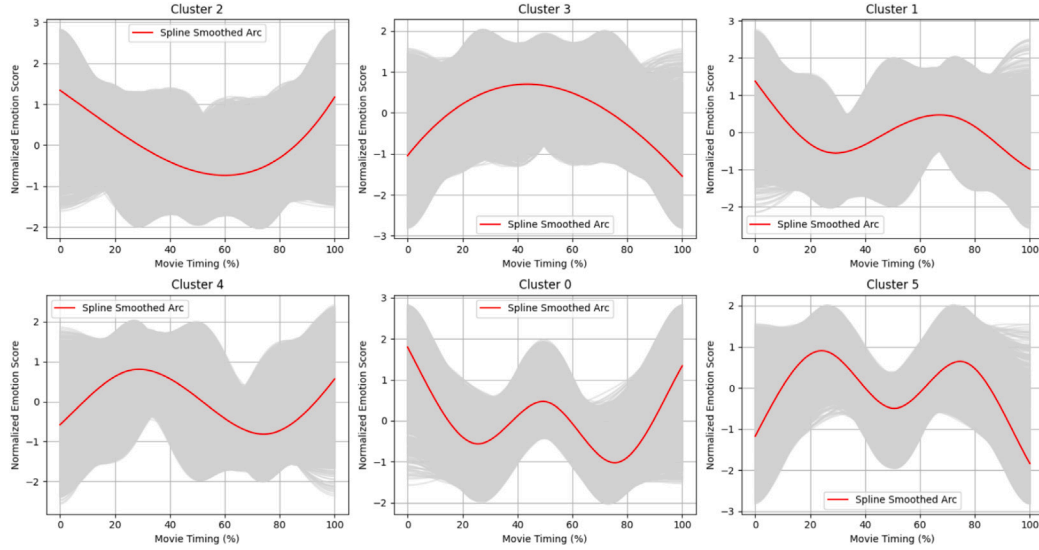


Fig. 3. Univariate Spline Plots of the Six different clusters. Image generated by authors.

The retained features include: *zero_crossings*, *skewness*, *negative_proportion*, *max_slope*, *range*, *shannon*, *rate_of_change_std*, *volatility*, *num_peaks*, *kurtosis*, *rate_of_change_mean*, *average*, *max*, *peak_to_peak*, *std*, *variance*, *general_positive_proportion*, *neutral_proportion*, *mean_peak_to_trough_slope*, *min*, *min_slope*, *average_slope*. A full list of the statistical features extracted can be found in [Appendix C](#).

3.3.3. Script embedding

The performance of the regression model was tested using these embedding models:

- *all-MiniLM-L6-v2* [45,46],
- *gte-modernbert-base* [47–49],
- *ModernBERT-embed-large* [50,51],
- *UAE-Large-V1* [52,53],
- *stella_en_400m_v5* [54,55],
- *jasper_en_vision_language_v1* [54],
- *stella_en_1.5B_v5* [54,56]

A fixed-size chunk of 512 tokens was used for models capable of handling such a number of input tokens and 256 tokens for the smaller *all-MiniLM-L6-v2*. Fig. 4 illustrates the chunking process adopted.

Instead of using a simple aggregation method, such as concatenating the embeddings (which would create unmanageably long vectors of varying lengths for different movies), a pooling strategy was used to unify them into a single, representative vector.

While popular pooling methods like *Mean Pooling* and *Max Pooling* exist, a custom weighted approach was taken to unifying embeddings.

Using a linearly weighted scale across the number of chunks, an increasing weight was allocated to embeddings as the movie progresses, with embeddings for the ending part of the movie being allocated the highest weighting. The rationale for this approach was based on the *Peak End Rule* principle which suggests that individuals' overall perception of an experience is heavily influenced by its most intense moment and its conclusion, rather than the experience in its entirety [57,58].

Keywords from the plot of each movie, were obtained from [59–61] as well as manually extracted using *KeyBERT* [62]. Saved as a list, the keywords were cleaned by using *WordNetLemmatizer* from NLTK [63] and removing stop words. The processed keywords were subsequently used to train a *Word2Vec* model with vector size 50, a window of 15, min_count of 1 and sg of 1. Each 50 dimensional vector was added to the main dataset as a feature.

3.3.4. Cast and crew inter-relationships

A significant amount of data is made available by IMDb with its regularly updated non-commercial datasets [34]. Besides extracting *star power*, *director power*, *writer power*, *producer power* and related experience features, given the historical data provided, it was also possible to analyse the interactions between cast and crew, through graph analysis.

Having created a dataset made up of individuals (being producers, writers, actors or directors) working on each movie, we used *rating* as weights to create a list of collaboration edges between individuals. Using *cugraph*, which is a library of graph algorithms which integrates into the RAPIDS data science ecosystem [64], we created the graph

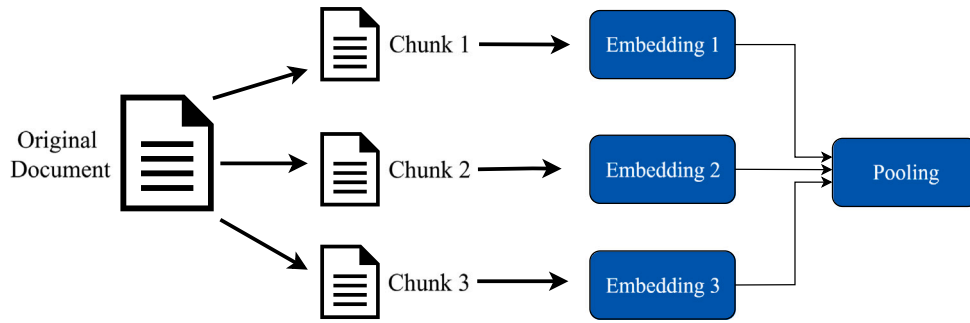


Fig. 4. The chunking process. Image generated by authors.

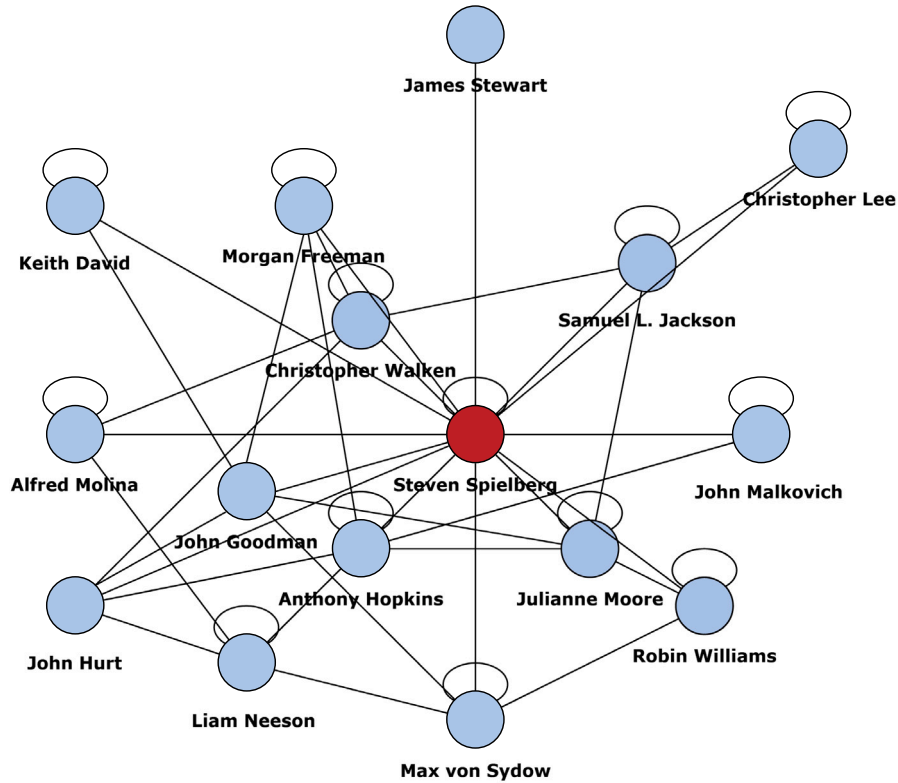


Fig. 5. An example of the top 15 connections with the director Steven Spielberg. Image generated by authors.

to incorporate the collaboration edges previously collated. *NetworkX*, which is a Python package for the creation, manipulation and study of the structure, dynamics and functions of complex networks [65] was subsequently used to better visualise the connections. A sample relationship graph for the director *Steven Spielberg* can be seen in Fig. 5.

Feature extraction was mainly conducted by analysing centrality measures. These included *degree centrality*, *betweenness centrality*, *PageRank*, *eigenvector centrality* and *average collaboration rating*.

3.3.5. Selection of regression models

Various regression models were tested for the experiments:

- XGBoost [66]
- LightGBM [67]
- Support Vector Machine [68,69]
- Lasso [69,70]
- CatBoost [71]
- ElasticNet [69,72]

Table 1

Classification of movies based on ratings.

Rating range	Classification
0.0 – 4.0	Low
4.1 – 7.0	Medium
7.1 and higher	High

- MLPRegressor [69]

These models were evaluated using R-squared (R^2), Mean Absolute Error (MAE), Root Mean Absolute Error (RMSE), Mean Absolute Percentage Error (MAPE) as well as a One-Away Score which gives a percentage of records from the test set where the predicted rating was within ± 1 of the actual rating. Accuracy and F1-Score were used for predicted categorisation of ratings as per Table 1.

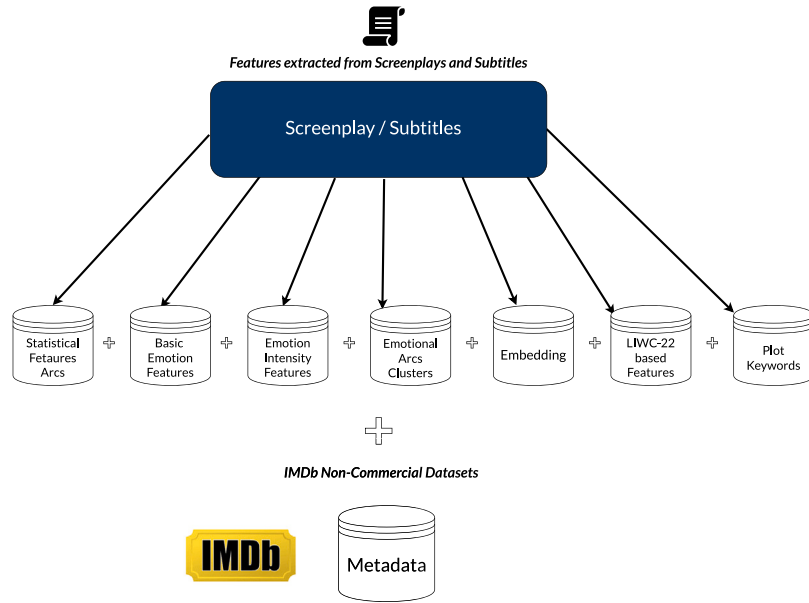


Fig. 6. Composition of training dataset. Image generated by authors.

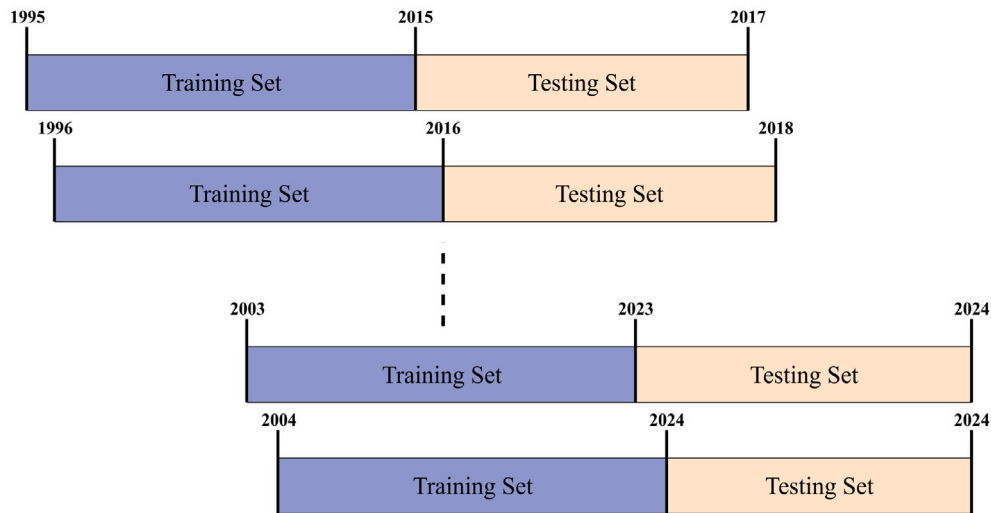


Fig. 7. Forward rolling window validation strategy. Image generated by authors.

4. Implementation

Feature data has been extracted from screenplays or subtitles, as the core source of information for this research. These have been concatenated to movie metadata sourced from IMDb as shown in Fig. 6.

4.1. Forward rolling window strategy

Building a training strategy for a movie rating prediction algorithm necessitated the consideration of the time factor. A forward rolling window validation strategy was adopted. A graphical representation of the strategy can be seen in Fig. 7.

An initial cut-off year of 2015 was selected, as the total amount of movies released before this year represent circa 80% of the whole dataset, with movies released 2015 up to 2024 representing circa 20% of the dataset. Training folds comprised movies released up to 20 years before the cut-off year and validation was carried out on movies released in the cut-off year and the subsequent 2 years. This training

and validation sequence is repeated by moving the cut-off year forward by one year until the last validation set being the movies released in 2024. Using this fixed-length training window (20 years) that shifts forward over time, model evaluation respects the chronological order of data to prevent data leakage.

Decisions at green lighting stage should be based on past data. This strategy ensures training data always comes from the past, with validation data being future based, relative to the training window.

A key component of the rolling window validation strategy was the need to update the training set with each forward shift. This update focused on the historical rating average for directors, writers, producers and main actors. Sourcing from a dictionary which stored the rolling average ratings of the previous five years of productions by each crew/cast member for each year from 2014 onwards, the training set features pertaining to these cast and crew members are automatically updated for the subsequent training fold. Similarly, the *experience* feature is also updated with the count of any new movies the respective cast or crew members would have been involved in the previous year.

Table 2

Comparison of training and testing sets for different fixed training window sizes.

Fixed training window size: 20 years		Fixed training window size: 10 years	
Training set	Testing set	Training set	Testing set
1994–2014	2015/2016/2017	2004–2014	2015/2016/2017
1995–2015	2016/2017/2018	2005–2015	2016/2017/2018
1996–2016	2017/2018/2019	2006–2016	2017/2018/2019
1997–2017	2018/2019/2020	2007–2017	2018/2019/2020
1998–2018	2019/2020/2021	2008–2018	2019/2020/2021
1999–2019	2020/2021/2022	2009–2019	2020/2021/2022
2000–2020	2021/2022/2023	2010–2020	2021/2022/2023
2001–2021	2022/2023/2024	2011–2021	2022/2023/2024
2002–2022	2023/2024	2012–2022	2023/2024
2003–2023	2024	2013–2023	2024

Table 3Performance Metrics using model with *crew_features*.

Model	Regression					Categorisation	
	R^2	RMSE	MAE	MAPE	One-Away	Accuracy	F1
Base	−0.0334	1.1244	0.8845	0.1848	0.6693	0.8236	0.7440
Fixed Window: 10							
all-MiniLM-L6-v2	0.4915	0.7875	0.6116	0.1225	0.8289	0.8332	0.7805
gte-modernbert-base	0.4936	0.7858	0.6087	0.1223	0.8264	0.8337	0.7772
UAE-Large-V1	0.4944	0.7847	0.6095	0.1215	0.8296	0.8321	0.7789
ModernBERT-embed-large	0.4979	0.7822	0.6093	0.1219	0.8316	0.8337	0.7812
jasper_en_vision_language_v1	0.5121	0.7709	0.5963	0.1200	0.8369	0.8373	0.7887
stella_en_400m_v5	0.5082	0.7742	0.5986	0.1202	0.8349	0.8371	0.7872
stella_en_1.5B_v5	0.5129	0.7703	0.5956	0.1200	0.8405	0.8362	0.7861
Fixed Window: 20							
all-MiniLM-L6-v2	0.4894	0.7892	0.6133	0.1230	0.8284	0.8339	0.7881
gte-modernbert-base	0.5031	0.7786	0.6034	0.1213	0.8349	0.8350	0.7855
UAE-Large-V1	0.5100	0.7727	0.6003	0.1200	0.8412	0.8326	0.7811
ModernBERT-embed-large	0.5127	0.7708	0.5984	0.1199	0.8366	0.8354	0.7881
jasper_en_vision_language_v1	0.5140	0.7696	0.5944	0.1197	0.8416	0.8362	0.7929
stella_en_400m_v5	0.5212	0.7640	0.5909	0.1188	0.8407	0.8350	0.7890
stella_en_1.5B_v5	0.5255	0.7604	0.5859	0.1183	0.8476	0.8375	0.7913

5. Evaluation

The analysis has been spread over 10 training and evaluation sets as in Table 2.

A base model is presented for comparative purposes, wherein the various test datasets are predicted using the mean rating of the training dataset. Two sets of comparative tables are presented, one which shows the results for the dataset containing crew related features (such as *combined_crew_power* and *combined_crew_experience*), and one without, thereby focusing principally on script related features. The results for the best performing model, SVR are being presented. With a dataset of a few thousand samples, the more complex models like XGBoost, LGBM, CatBoost and MLPs tend to overfit, while SVR reported a stable performance as a result of adequately scaled features and strong regularisation. Deep learning models can have hundreds of thousands, if not millions of parameters, allowing them to learn complex, high-dimensional functions, with the risk of capturing not only the underlying patterns of the training data but also its noise. Having vast amounts of data records mitigates this risk, enabling deep learning models to generalise better over unseen data. With the available records, collating a large enough dataset was not possible, resulting in overfitting of deep learning models.

The results in Tables 3 and 4 give a strong indication that *ceteris paribus*, the choice of embedding model for the script influences the metrics, with larger models (having more parameters) yielding better scores. Reviewing the results in Table 3 using the best performing embedding model, one can note an R^2 of 0.5255. This indicates that approximately 52.55% of the variability in the IMDb ratings is explained by the model. Many factors that influence a movie's success (such as audience reception or cultural timing) are unpredictable, so

achieving this level of explanatory power at such an early stage may be considered significant.

Comparing the MAE and the RMSE one can note a slight disparity in the values, with RMSE slightly higher at 0.7604 compared to the MAE of 0.5859. While the latter provides a measure of the average magnitude of errors (irrespective of direction) and hence can be directly interpreted as the predictions being, on average, 0.5859 units away from the actual IMDb rating, the higher RMSE value suggests that there might be some larger errors that are affecting the RMSE more significantly. For a producer, this means if the model forecasts a rating of 7.2, the actual rating is very likely to land between 6.6 and 7.8. To analyse prediction errors, a scatter plot was generated, displaying predicted ratings against actual ratings. The size of each point reflects the magnitude of the absolute error. Fig. 8 illustrates the model's performance on the validation set, which consists of movies released in 2016–2018, while the model was trained on movies from 1995–2015.

The MAPE of 0.1183 or 11.83% means that on average, the model's predictions deviate from the actual IMDb ratings by 11.83%. The one-way accuracy metric indicates that the model was, on average (across all training folds), able to predict the IMDb rating within one point of the actual rating for 84.76% of the movies in the validation sets. In practical terms, this makes the model a reliable 'early compass' for audience reception.

For the classification exercise, ratings were binned into 3 categories, [0–4], [4.1–7], [7.1–10]. The model managed to correctly place into the correct bin 83.75% of the movies in the validation sets, while achieving an F1-score of 0.7913. Having an imbalanced dataset, the F1-score was of particular importance as it provides a balanced measure of the model's performance.

An in-depth analysis of the validation dataset composition was conducted to inspect whether the same metrics were achieved for full

Table 4
Performance Metrics using model without *crew_features*.

Model	Regression					Categorisation	
	R^2	RMSE	MAE	MAPE	One-Away	Accuracy	F1
Base	-0.0334	1.1244	0.8845	0.1848	0.6693	0.8236	0.7440
Fixed Window: 10							
all-MiniLM-L6-v2	0.4146	0.8446	0.6581	0.1301	0.8000	0.8295	0.7715
gte-modernbert-base	0.4229	0.8385	0.6525	0.1302	0.8015	0.8308	0.7753
UAE-Large-V1	0.4241	0.8374	0.6527	0.1289	0.8051	0.8291	0.7764
ModernBERT-embed-large	0.4321	0.8316	0.6481	0.1287	0.8046	0.8309	0.7783
jasper_en_vision_language_v1	0.4476	0.8200	0.6352	0.1266	0.8170	0.8353	0.7859
stella_en_400m_v5	0.4510	0.8175	0.6334	0.1265	0.8164	0.8324	0.7819
stella_en_1.5B_v5	0.4588	0.8115	0.6278	0.1259	0.8212	0.8391	0.7906
Fixed Window: 20							
all-MiniLM-L6-v2	0.4242	0.8379	0.6538	0.1291	0.8002	0.8289	0.7750
gte-modernbert-base	0.4283	0.8345	0.6487	0.1293	0.8046	0.8310	0.7780
UAE-Large-V1	0.4375	0.8274	0.6444	0.1273	0.8098	0.8304	0.7780
ModernBERT-embed-large	0.4426	0.8240	0.6407	0.1271	0.8100	0.8310	0.7810
jasper_en_vision_language_v1	0.4566	0.8135	0.6292	0.1254	0.8184	0.8337	0.7858
stella_en_400m_v5	0.4600	0.8108	0.6287	0.1253	0.8225	0.8318	0.7842
stella_en_1.5B_v5	0.4681	0.8046	0.6216	0.1248	0.8235	0.8389	0.7932

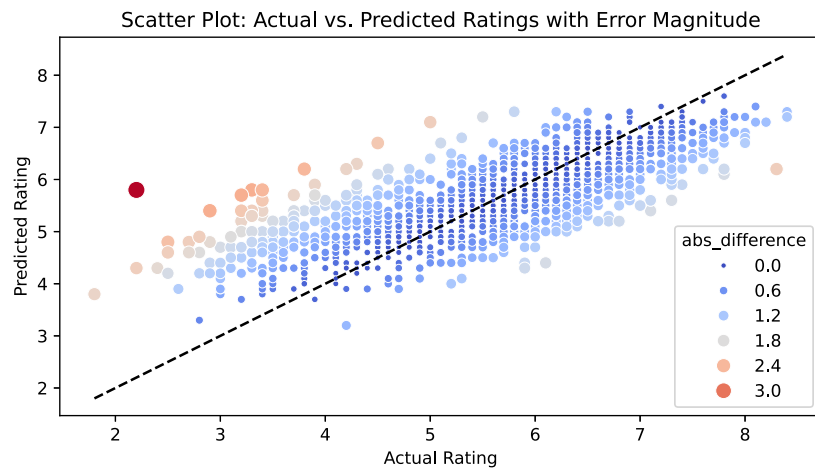


Fig. 8. Scatter Plot — Actual vs Predicted ratings - Cut-off year 2015. Image generated by authors.

Table 5
Mean of metrics per type of script.

Type	Mean				
	R^2	RMSE	MAE	MAPE	One-Away
Overall	0.5255	0.7604	0.5859	0.1183	0.8476
Full Screenplays	0.2898	0.6658	0.5162	0.0814	0.9048
Subtitles	0.4795	0.7699	0.5936	0.1224	0.8412

screenplays *versus* subtitles. The results are summarised in [Tables 5](#) and [6](#). Averages and Standard Deviation were calculated on a weighted basis. Of particular interest was the fact that with the exception of the R^2 metric, all other metrics were significantly better for full screenplays when compared with those of subtitles. Values indicate that predictions for full screenplays tend to have lower absolute errors. At 90.48%, predictions for full screenplays are also more probable to be within one unit of the actual rating. Observations from the standard deviations indicate that while having lower absolute errors, predictions on full screenplays have the most inconsistent performance. This could be the result of the smaller proportion in both training and validation sets.

A visualisation of the metrics over the respective validation time frames is shown in [Fig. 9](#). The volatility of the full screenplay predictions can be clearly noticed, where predictions on subtitle based scripts tend to be more stable over the validation time-frames.

Table 6
Standard deviation of metrics per type of script.

Type	Standard deviation				
	R^2	RMSE	MAE	MAPE	One-Away
Overall	0.0388	0.0150	0.0110	0.0031	0.0080
Full Screenplays	0.0723	0.0570	0.0415	0.0082	0.0351
Subtitles	0.0360	0.0176	0.0138	0.0038	0.0093

An additional visualisation is being presented in [Fig. 10](#) to show the comparison between training and testing results for the compound (full screenplays and subtitles) version of the dataset, across the evaluation time-frames.

5.1. Comparative analysis

Benchmarking a model's results against those of other architectures is an important element to determine whether any improvement has been achieved. As the data for this paper was collated from scratch, no direct comparison can be performed on the results.

The evaluation of the model presented in this paper however reveals its competitive edge over existing methodologies in movie rating prediction. In comparison to Ning et al.'s work [\[73\]](#), which proposed a generative convolutional neural network leveraging pre-production movie features like genres, budget, cast, director, and plot information,

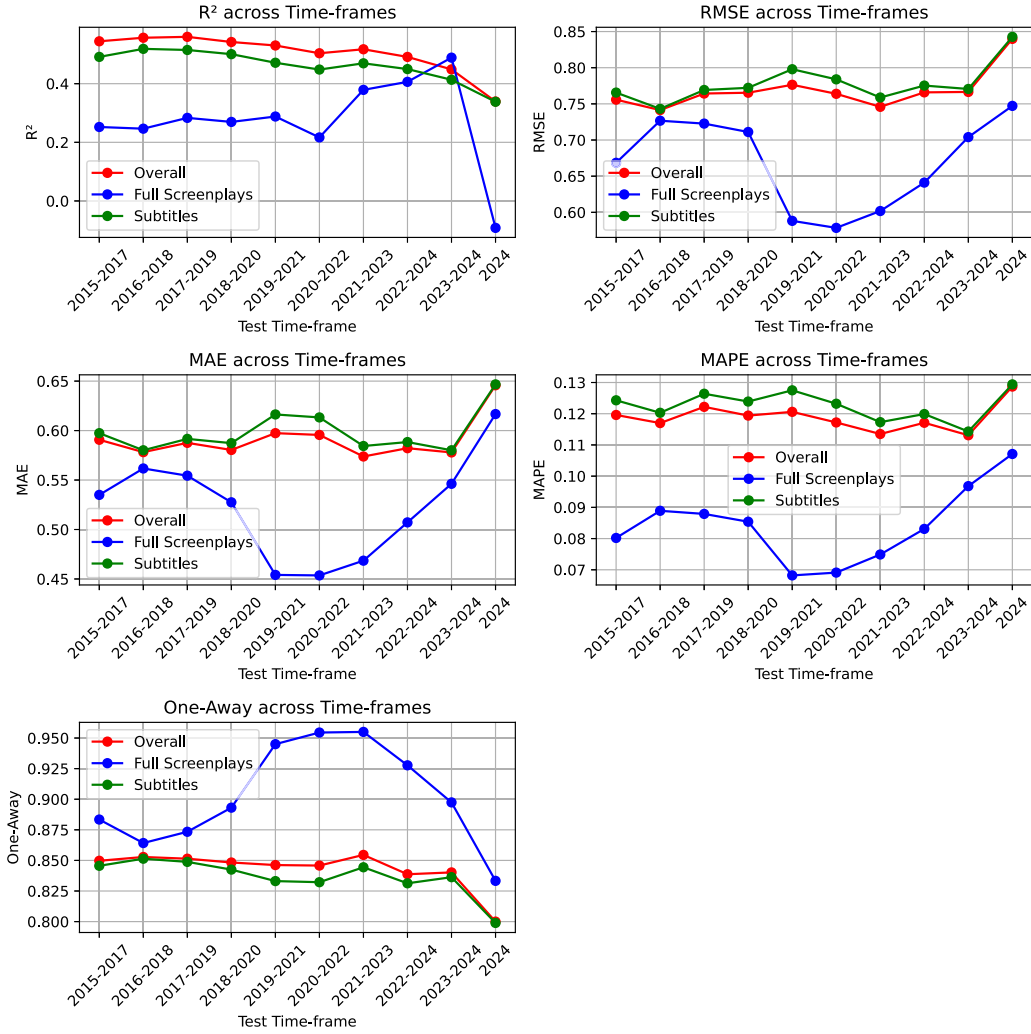


Fig. 9. Metrics across time frames. Image generated by authors.

this model demonstrates improved performance. Their model, trained on 1219 movies released between 2003 and 2013 and tested on 252 movies released after 2013, achieved a MSE of 0.628 and a one-away accuracy of 85.3%. In contrast, this SVR model with a fixed 20-year window achieved a lower MSE of 0.5782 (derived from an RMSE of 0.7604) and a slightly lower one-away accuracy of 84.76%.

A study of 1687 Indian Movies released in or after 2010 with features comprising average actor, actress, composer, director, producer and writer ratings, together with the genre and movie runtime, used a Random Forest model to achieve an RMSE of 1.2276 when predicting the average online viewer rating from IMDb [74]. Our model achieved an RMSE of 0.7604.

Similarly, Abarja et al. [7] employed a Convolutional Neural Network (CNN) on a larger dataset of 3819 training movies (2000–2013) and 498 testing movies (after 2013), incorporating metadata, historical, topical, and categorical features, but reported a higher MSE of 1.20 and a Mean Absolute Error (MAE) of 0.83. Furthermore, our model's MAE of 0.5859 significantly outperforms the MAE (referred to as Prediction Absolute Error, PAE) of 0.6973 obtained by Hsu [6] in a study using IMDb data without a strict chronological split between training and testing sets, highlighting the importance of the temporal validation approach used in this research.

A dissertation based on Kaggle's IMDb Movies Dataset achieved an R^2 of 0.335 and a RMSE of 0.859 using a random forest algorithm [75]. Finally, Zabaleta's dissertation [76], which utilised features such as release year, runtime, 22 genres, 70 regions, and over 3000 artists,

reported an MAE of 0.684, again higher than the MAE achieved by our model. These comparisons highlight the effectiveness of this paper's regression model in predicting movie ratings.

6. Conclusion

This study aimed to build a predictive model for movie ratings by analysing a custom dataset of scripts and subtitles. The model, which incorporated linguistic features, script embeddings, and crew-related data, achieved a robust average performance with an R^2 of 0.5255 and a one-away score of 0.8476. The analysis successfully demonstrated that features extracted from scripts and crew information have a significant correlation with audience reception.

The primary novelty of this research lies in its multi-faceted approach to feature engineering. We move beyond traditional single-source analysis by introducing a model that integrates a rich array of features, including fine-grained linguistic metrics from film scripts, high-dimensional text embeddings, emotional content and structured crew-related data. By combining these disparate, yet complementary, data types, our methodology establishes a more sophisticated framework for predicting audience reception. This innovative feature combination represents a significant advancement over existing literature, highlighting a path towards more accurate and comprehensive predictive modelling in the domain of film analytics.

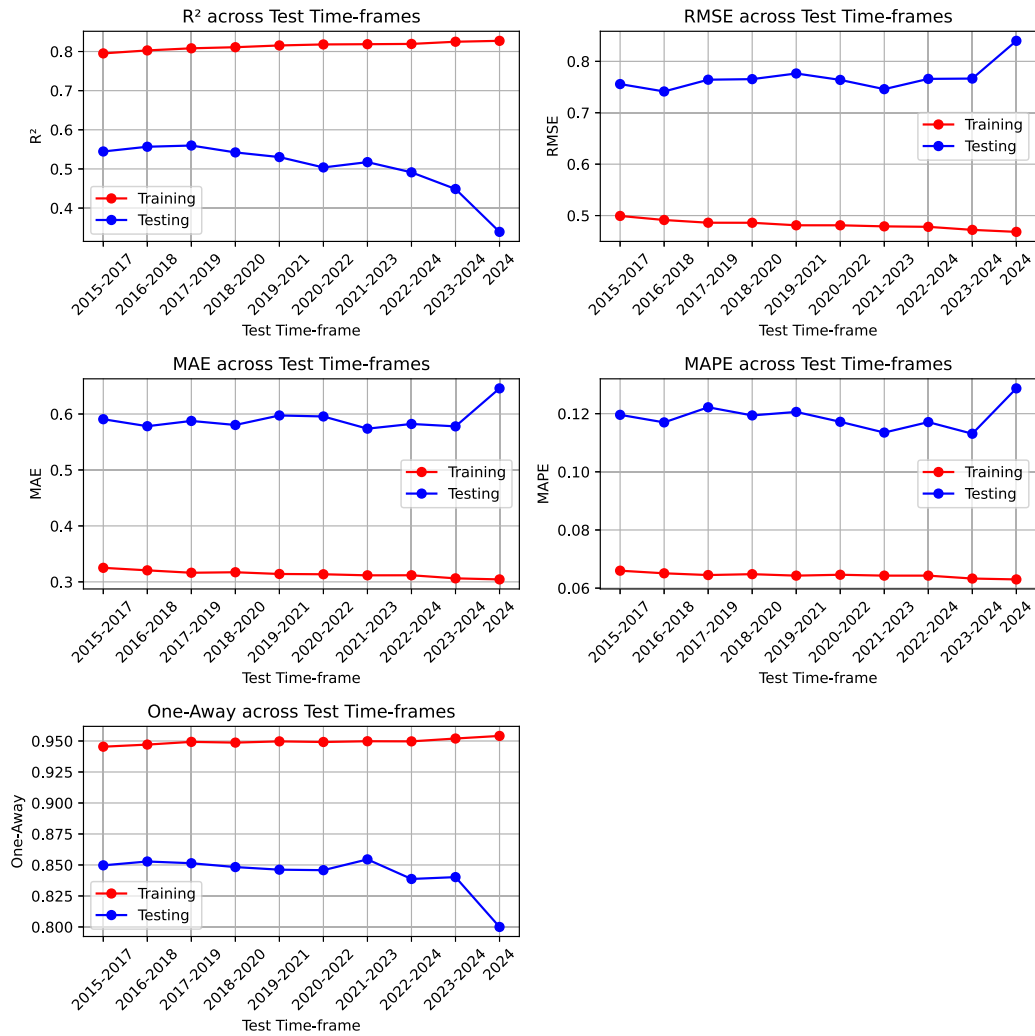


Fig. 10. Training and Testing Metrics across time frames. Image generated by authors.

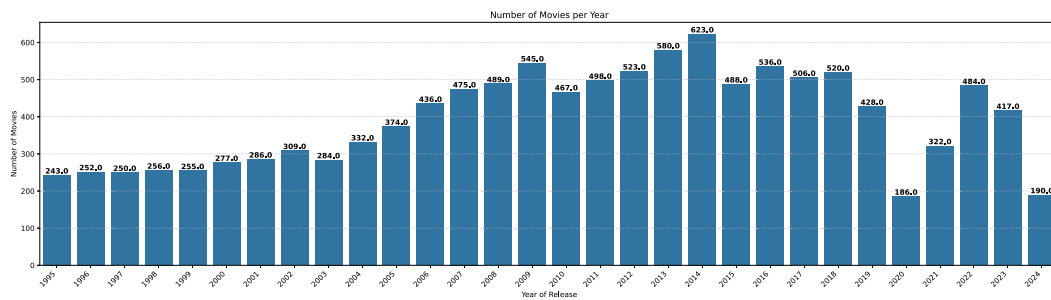


Fig. A.11. Number of movies — 1995 onwards. Image generated by authors.

6.1. Practical implications

The methods in this research provide a practical application that can be used by producers in real world decision making. Movies predicted to land in the 7–8 rating range could be considered safer bets for broader audience appeal, while those projected below 5.5 may carry substantial risk, unless other business factors such as a franchise potential or a major star actor could outweigh the quality signal. The model can also provide an objective early-stage metric to help secure a portfolio of projects which include a balance of high-risk/high-reward

projects with safer bets. Besides providing a reasonably confident prediction on the actual rating it also opens up the possibility of researching optimal crew and cast combinations for different genres of movies.

From an investor point of view, this data-driven estimate of how audiences are likely to respond, reduces reliance on pure intuition or reputation. Multiple projects can be compared on their predicted audience rating, helping the allocation of capital towards projects with the most potential. It can also help identify niche projects that might

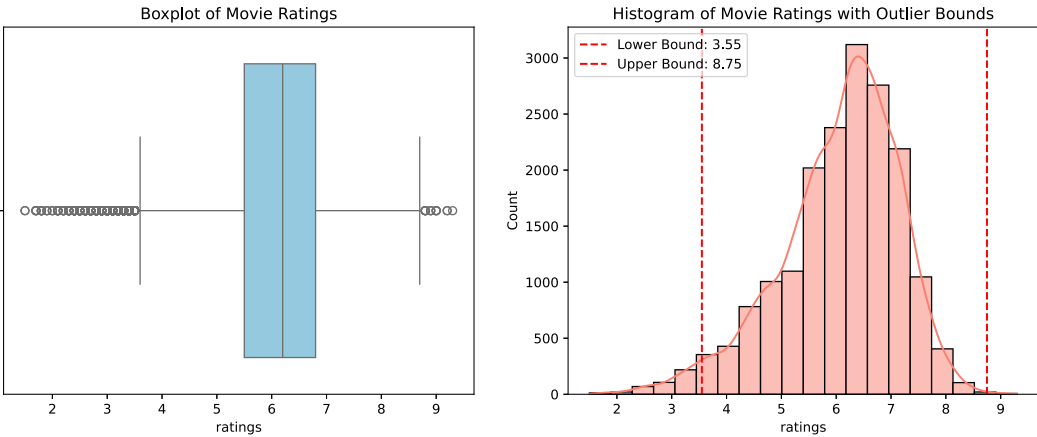


Fig. A.12. Analysis of ratings in Dataset. Image generated by authors.

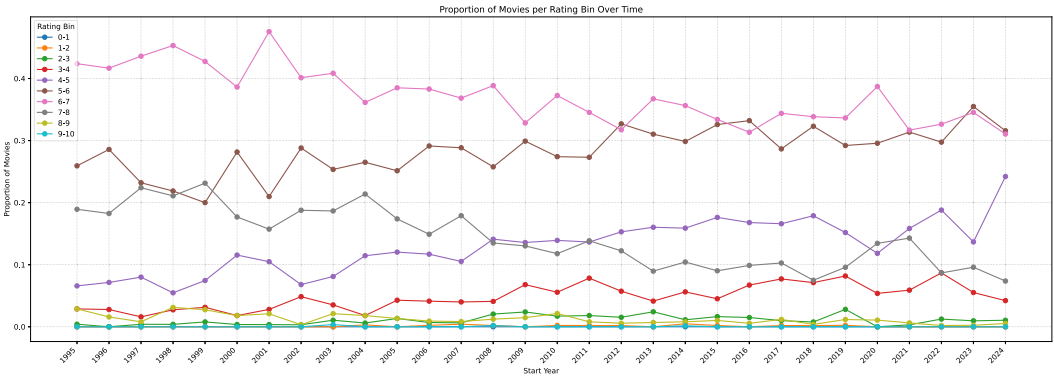


Fig. A.13. Analysis of the distribution of ratings in dataset across the years. Image generated by authors.

Table B.7
LIWC extracted features from movie scripts.

WC	Analytic	Clout	Authentic	Tone
WPS	BigWords	Dic	Linguistic	function
pronoun	ppron	i	we	you
shehe	they	ipron	det	article
number	auxverb	adverb	conj	negate
verb	adj	quantity	Drives	affiliation
achieve	power	Cognition	allnone	cogproc
insight	cause	discrep	tentat	differ
memory	Affect	tone_pos	tone_neg	emotion
emo_pos	emo_neg	emo_anx	emo_sad	swear
Social	socbehav	prosocial	polite	conflict
moral	comm	socrefs	family	female
male	Culture	politic	ethnicity	tech
Lifestyle	leisure	home	work	money
relig	Physical	illness	wellness	mental
substances	sexual	food	death	need
want	acquire	lack	fulfil	fatigue
reward	risk	curiosity	allure	Perception
attention	space	visual	auditory	feeling
time	focuspast	focuspresent	Conversation	netspeak
assent	nonflu	filler	AllPunc	Period
Comma	QMark	Exclam	Apostro	Narrativity_PlotProg
Narrativity_CogTension	Staging_10	PlotProg_10	CogTension_1	CogTension_8

have a lower predicted rating but which show potential if matched with a good distribution model or marketing strategy.

6.2. Limitations and future work

The research faced key limitations related to data and infrastructure. The availability of full screenplays for this research was restricted.

Results have indicated that while being more volatile, predictions on full screenplays outperform those on subtitle based screenplays. The added contextual scene information found in full screenplays enables a richer interpretation and understanding of specific movie features by models and having a larger database of full screenplays would provide researchers with the possibility to reduce this volatility and train a better generalised model. A larger dataset combined with stronger

Table C.8

List of features extracted from emotional arcs with descriptions.

Feature	Description
<i>Average</i>	The mean value of the emotional arc over time.
<i>Variance</i>	A measure of the spread or dispersion of the emotional arc values.
<i>Standard Deviation</i>	The square root of variance, indicating the amount of variation in the emotional arc.
<i>Minimum Value</i>	The smallest value in the emotional arc.
<i>Maximum Value</i>	The largest value in the emotional arc.
<i>Range</i>	The difference between the maximum and minimum values in the emotional arc.
<i>Median</i>	The middle value of the emotional arc when sorted in ascending order.
<i>Kurtosis</i>	A measure of the “tailedness” of the emotional arc distribution.
<i>Skewness</i>	A measure of the asymmetry of the emotional arc distribution.
<i>Average Slope</i>	The mean rate of change between consecutive points in the emotional arc.
<i>Maximum Slope</i>	The steepest positive rate of change in the emotional arc.
<i>Minimum Slope</i>	The steepest negative rate of change in the emotional arc.
<i>Total Curvature</i>	The cumulative curvature of the emotional arc, indicating how much the arc bends.
<i>Mean Trough to Peak Slope</i>	The average slope between troughs and subsequent peaks in the emotional arc.
<i>Permutation Entropy</i>	A measure of the complexity of the emotional arc based on ordinal patterns.
<i>Shannon Entropy</i>	A measure of the unpredictability or randomness in the emotional arc.
<i>Autocorrelation</i>	A measure of the similarity between the emotional arc and a time-lagged version of itself.
<i>Mean Rate of Change</i>	The average rate at which the emotional arc changes over time.
<i>Standard Deviation of Rate of Change</i>	The variability in the rate of change of the emotional arc.
<i>Zero Crossings</i>	The number of times the emotional arc crosses the zero line.
<i>Positive Proportion</i>	The proportion of the emotional arc that lies above the zero line.
<i>Negative Proportion</i>	The proportion of the emotional arc that lies below the zero line.
<i>Neutral Proportion</i>	The proportion of the emotional arc that lies close to the zero line.
<i>Distance Peak to Peak</i>	The average distance between consecutive peaks in the emotional arc.
<i>Volatility</i>	A measure of the variability or fluctuations in the emotional arc.
<i>Mean Peak to Trough Slope</i>	The average slope between peaks and subsequent troughs in the emotional arc.
<i>Number of Peaks</i>	The total number of local maxima in the emotional arc.
<i>Number of Troughs</i>	The total number of local minima in the emotional arc.

regularisation would also assist with reducing overfitting in the training of deep learning models.

The models used could have been further optimised. The high-dimensional dataset and lack of comprehensive hyperparameter tuning, feature selection, and testing of different training windows limited the models’ performance.

Technical infrastructure prohibited us from carrying out detailed feature importance analysis. Besides individual features, categories of features such as statistical features, emotion related features, embeddings and linguistic features can be ranked in order of importance by using techniques such as permutation importance or Shapley values.

The study focused on correlations rather than causal effects, primarily due to the lack of access to critical variables like marketing budgets and distribution strategies. Additionally, the use of large language models and hyper-parameter optimisation was restricted by available resources, which could have led to a more refined and accurate model. Future work should aim to incorporate more data on causal factors and utilise advanced modelling techniques to evolve this tool into an even more robust decision-support system for producers.

CRediT authorship contribution statement

Karl Cini: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **John Abela:** Writing – review & editing, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Chat GPT-5 to format LaTeX tables to meet the specific requirements of the journal. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Dataset composition

In Section 3.1 we explain that 2785 full length screenplays were obtained from different sources. An additional 32,077 titles were sourced from three other repositories to bring the total of records in the dataset to 34,862 before filtering.

To obtain a valid representation of movies to train our models we filtered down the dataset based on several factors including removing titles with less than 1000 registered votes on IMDb and movies whose original language was not English. We also removed documentaries, music videos and biographies. Finally we also removed duplicates to retain 18,149 titles for model training.

Training of the models was carried out using specific time windows, as discussed in Section 4.1, starting from 1995 onwards.

The number of movies representing each year (of release) can be seen in Fig. A.11.

A box plot and a histogram of the ratings distribution is shown in Fig. A.12. Outlier bounds were set at $1.5 \times$ the Interquartile Range (IQR).

The distribution of ratings of movies across the years is also stable as indicated in Fig. A.13.

Appendix B. LIWC extracted features

In this Appendix we list the features that have been retained from the LIWC extraction process after using Recursive Feature Elimination. In total, 115 out of the 151 dimensions were retained for concatenation to the main metadata vector (see Table B.7).

Appendix C. Statistical features — emotion analysis

In this Appendix we give a more detailed explanation of the statistical features extracted from the emotion analysis in Section 3.3.2 (see Table C.8).

Appendix D. Description of data usage

The data for this research includes a collection of movie screenplays. These screenplays served as the source material for computational analysis aimed at exploring the relationship between script characteristics and movie audience reception, specifically as reflected in IMDb user ratings. This is a non-commercial, academic endeavour.

The methodology involved processing the full text of each screenplay to extract quantifiable features. This process included:

- **Linguistic Feature Extraction:** Analysing the text to identify and quantify various linguistic elements.
- **Narrative Structure Analysis:** Examining the script structure to derive metrics related to narrative flow and organisation.
- **Emotional Trajectory Analysis:** Assessing the emotional content within the dialogue and descriptions throughout the script.
- **Script Embedding:** Transforming the entire script text into vector representations that capture semantic and stylistic information.

The screenplays were utilised strictly as data inputs for these automated analytical techniques. The output of this processing is a dataset of numerical features and vector representations derived from the scripts.

Crucially, this research does not involve the reproduction, distribution, or display of the original movie screenplays, subtitles or any significant portions thereof in a human-readable format. No dialogue, scene descriptions, or other textual content from the scripts will be quoted or presented for the reader to consume as they would the original work. Furthermore, no audio or visual components of the movies are used or included. The analysis operates on the computational representations of the scripts, and the research presents the findings derived from this data and the resulting predictive model. The use of the screenplays is confined to their role as data within this analytical research process.

Data availability

Some of the data can be made available upon request.

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